Lecture 20: Linear Regression I

Heather Mattie, PhD

November 6, 2025

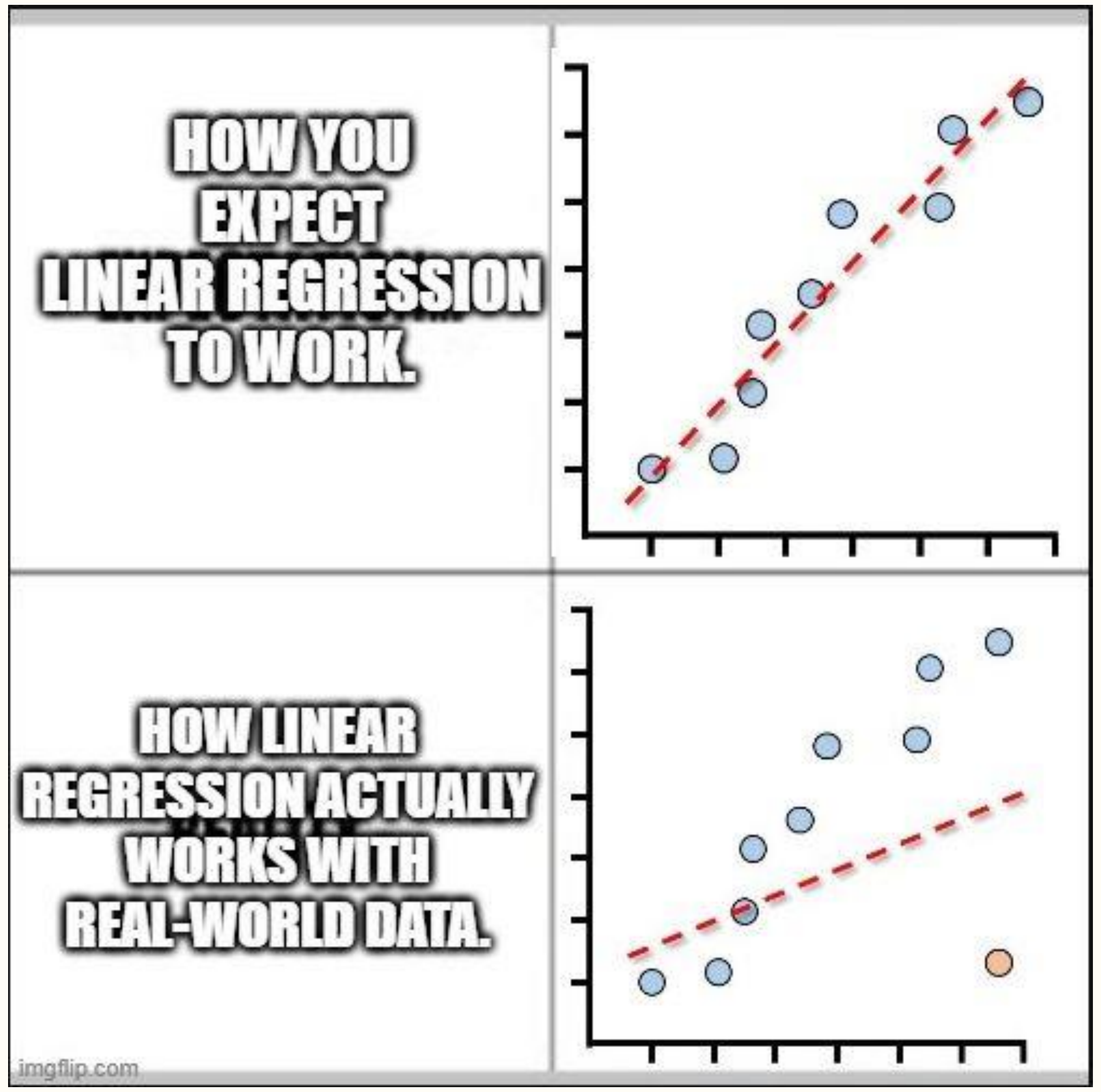
Recipe of the Day!

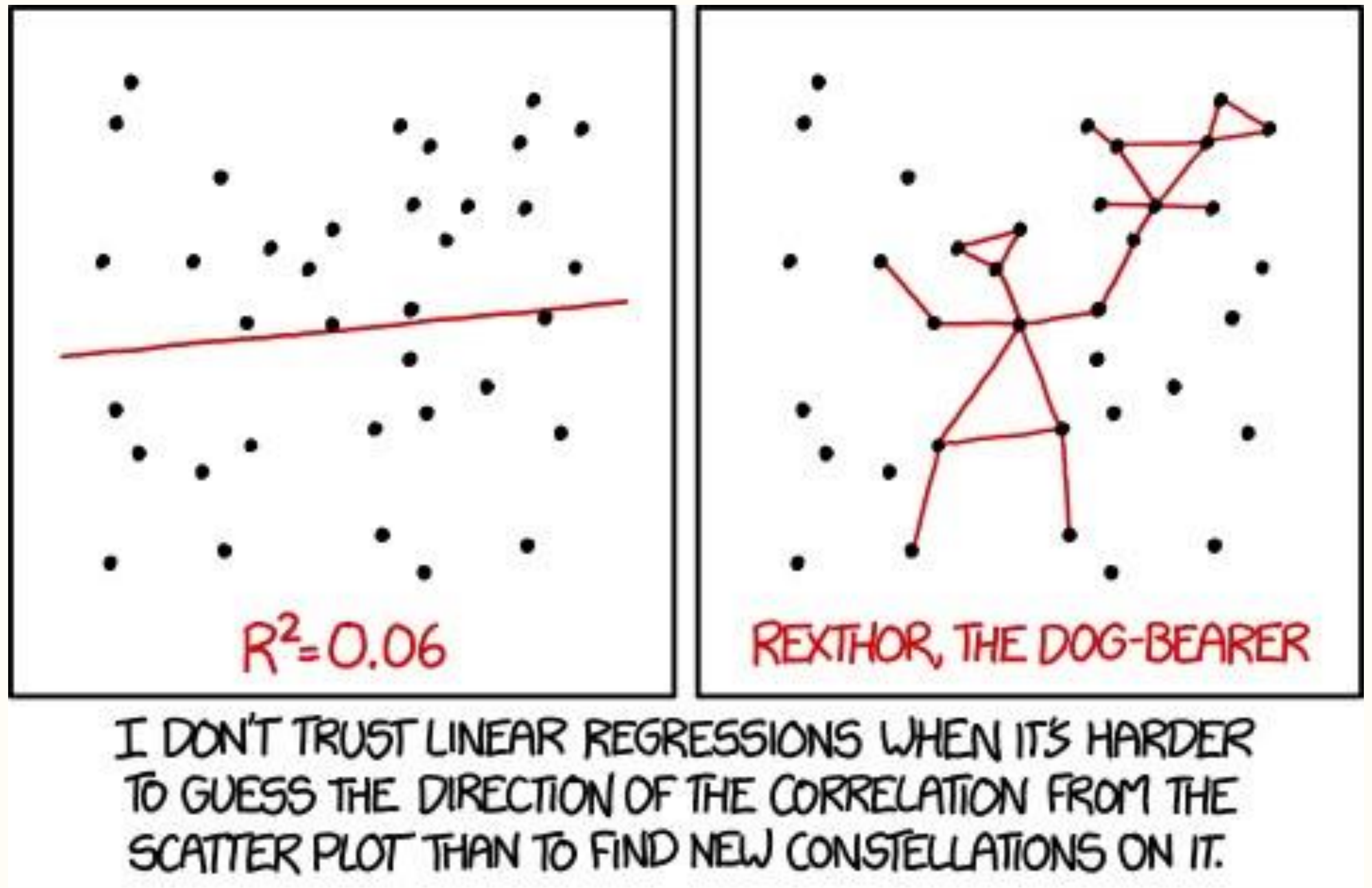
[Shepherd's Pie]

[Franklin Park Zoo Lights]

Announcements

- No lab this week
- Midterm due **Friday, November 7**
  - No late days may be used
- No class November 11, 25, 27





Coding question of the day

The data frame we created on Tuesday (11/4) contains a column called `cases`. The values of this column are the cumulative number of reported cases in a particular country on a particular day. Use the `lag` function to add a new column to the data frame that is the number of new cases for each day. For example, if there were 10 reported cases on January 22nd and 15 reported cases on January 23rd, the number of new cases would be 5 (15-10). Name this column `new_cases`.

Bonus challenge: create a new column that contains the seven-day rolling average of new cases. Name this column `new_cases_7dayavg`. I would suggest using the `rollmean` function from the `zoo` package.

This code is from the 11/4 coding question of the day.

```
library(readr)
library(dplyr)
library(tidyr)
library(lubridate)

data <- read_csv("https://raw.githubusercontent.com/CSSEGISandData/COVID-19/refs/heads/master/archived_data/archived_time_series/time_series_19-covid-Recovered_archived_0325.csv")

data_long <- data %>%
  pivot_longer("1/22/20":"3/23/20", names_to = "date", values_to = "cases") %>%
  mutate(date = mdy(date))

data_long_country <- data_long %>%
  rename(country = "Country/Region") %>%
  group_by(country, date) %>%
  summarize(cases = sum(cases))
```

Linear Regression I

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Linear Regression I Roadmap

1. Correlation
2. Simple linear regression
3. Evaluation of model assumptions

Linear Regression I

Correlation

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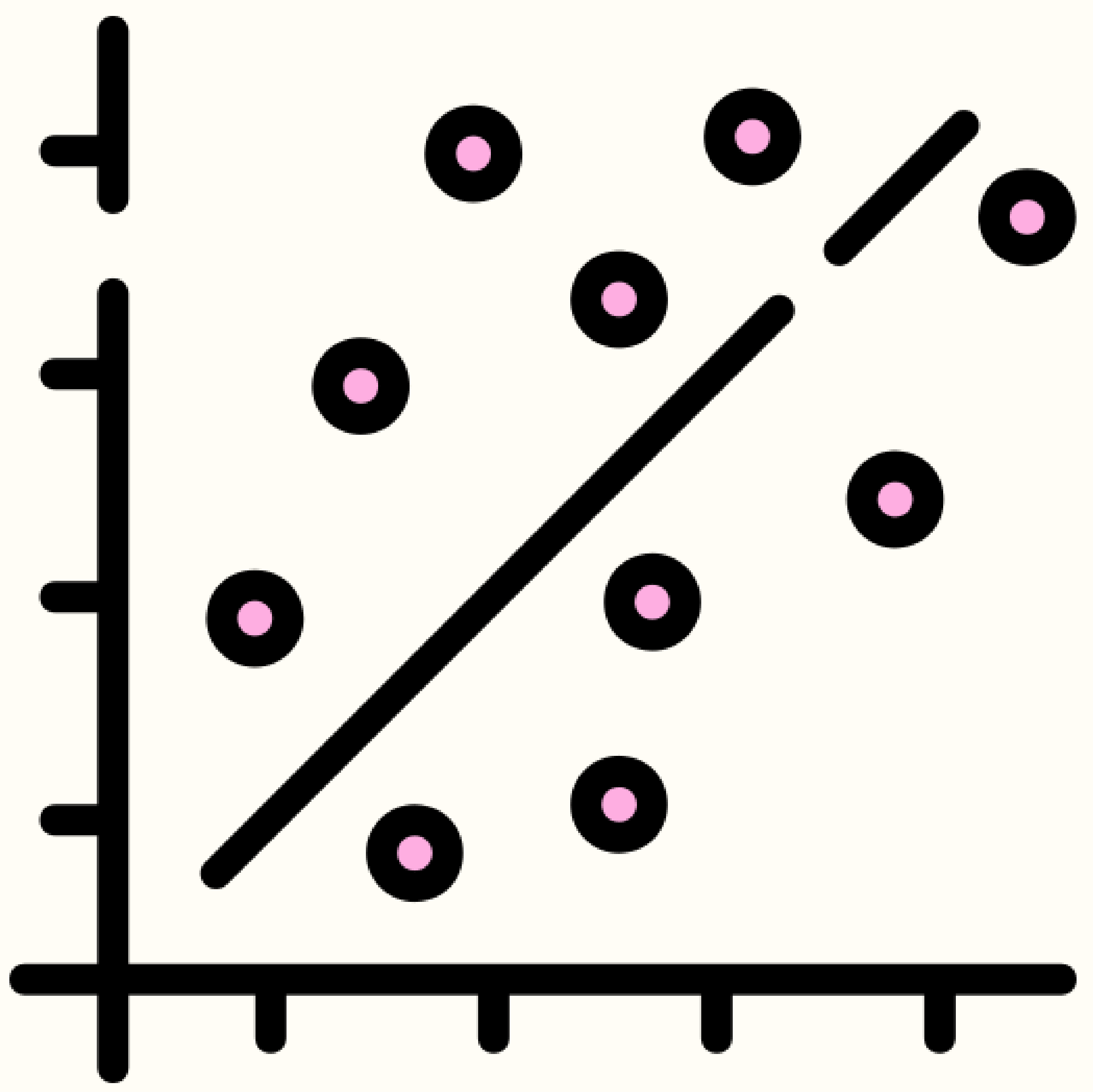
Correlation

Correlation quantifies the **linear** relationship between two continuous variables

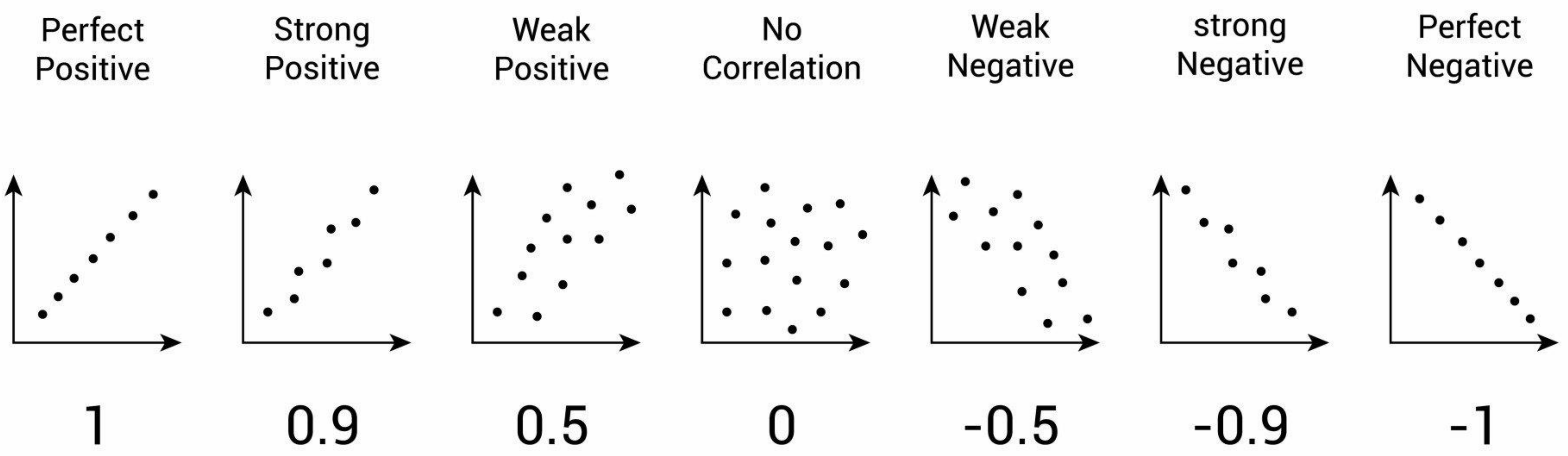
- Can measure the magnitude of the association between the two variables
- Scatterplots help visualize the relationship
- Two correlation estimates: Pearson correlation coefficient and Spearman rank correlation coefficient



Correlation

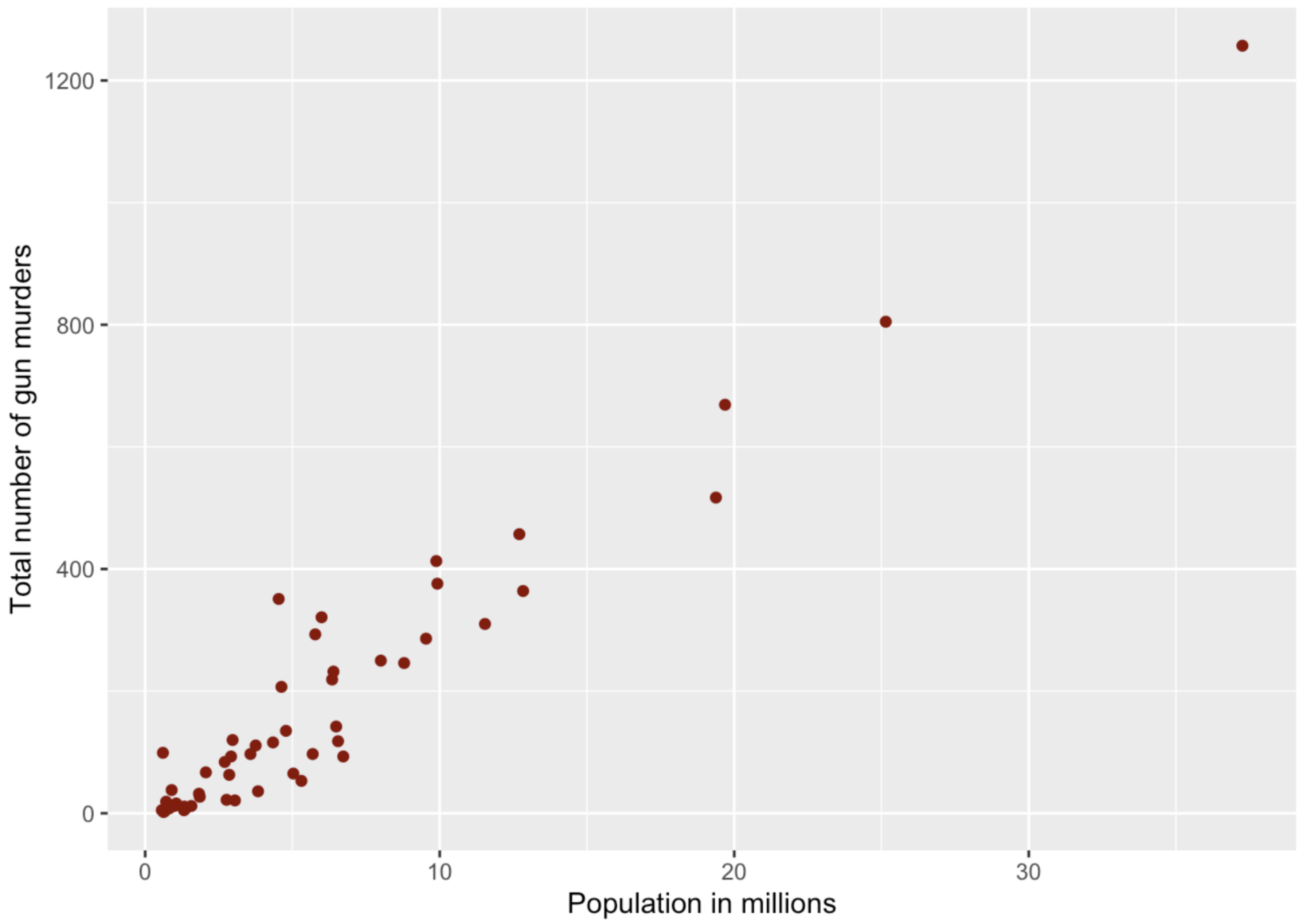
Correlation ranges from -1 to 1 and has no units of measurement

- Values closer to 1 or -1 represent a stronger linear relationship
- A value of 0 means no **linear** relationship



Example: gun murders

```
murders %>%
  ggplot(aes(population/10^6, total)) +
  geom_point(color = "darkred") +
  xlab("Population in millions") +
  ylab("Total number of gun murders")
```



Pearson correlation coefficient

The Pearson correlation coefficient (*r*) is a point estimate of ρ (the population correlation)

Assumption:

- Both variables are approximately normally distributed

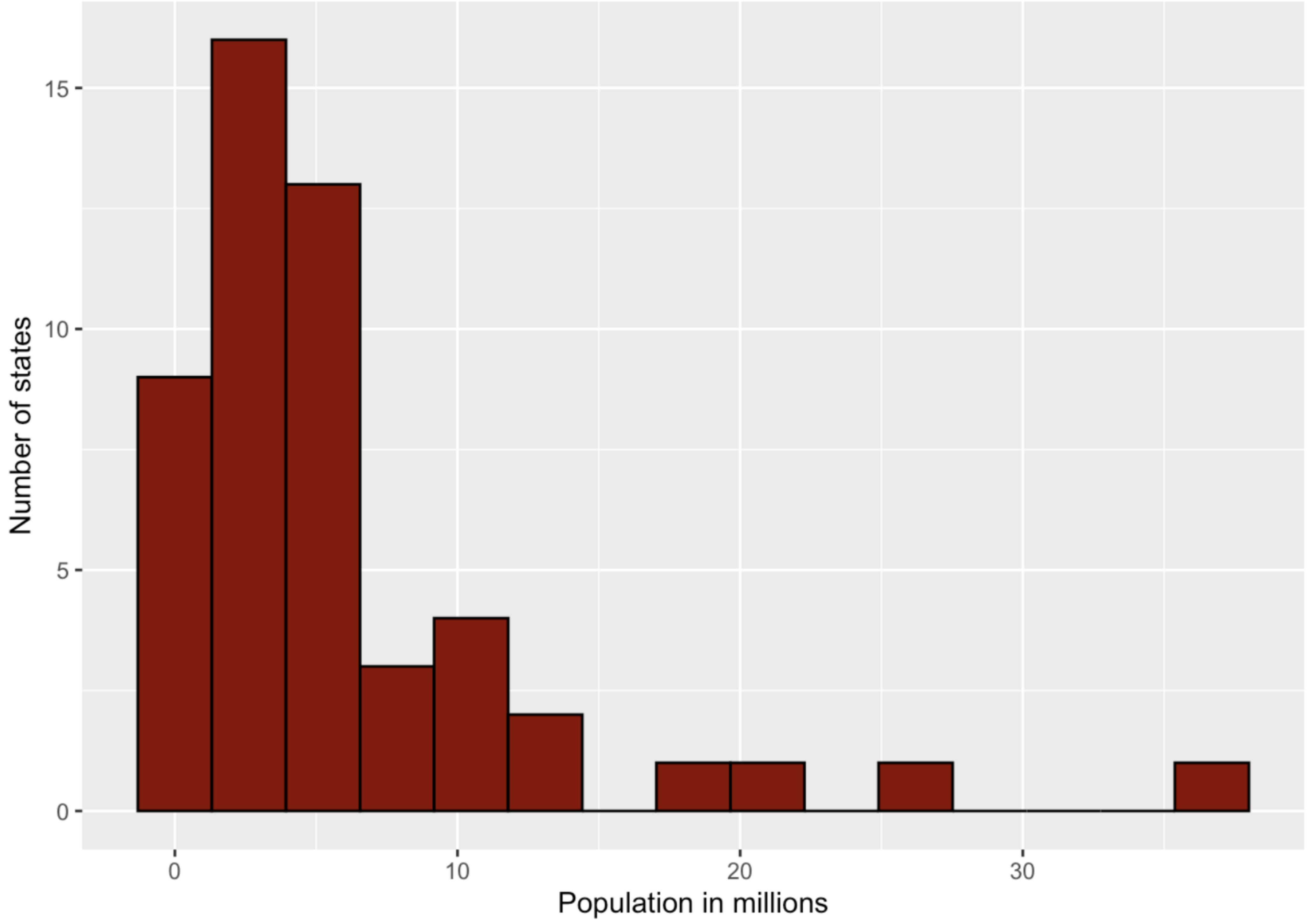
Pearson correlation coefficient

The Pearson correlation coefficient (r) is a point estimate of ρ (the population correlation)

Assumption:

- Both variables are approximately normally distributed

```r
murders %>%
  ggplot(aes(population/10^6)) +
  geom_histogram(color = "black", fill = "darkred", bins = 15) +
  xlab("Population in millions") +
  ylab("Number of states")
```



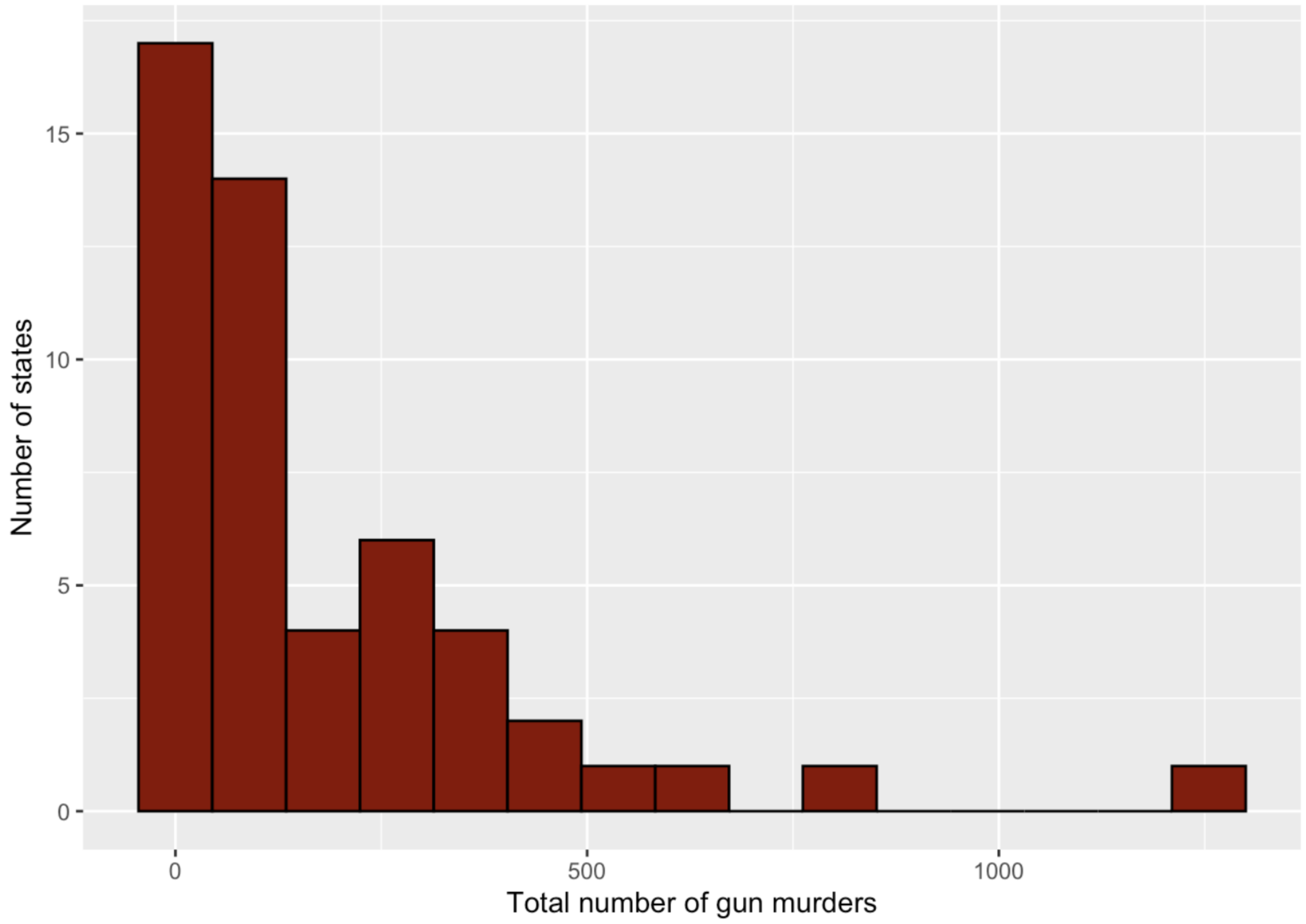
Pearson correlation coefficient

The Pearson correlation coefficient (r) is a point estimate of ρ (the population correlation)

Assumption:

- Both variables are approximately normally distributed

```r
murders %>%
  ggplot(aes(total)) +
  geom_histogram(color = "black", fill = "darkred", bins = 15) +
  xlab("Total number of gun murders") +
  ylab("Number of states")
```



Pearson correlation coefficient

The Pearson correlation coefficient (r) is a point estimate of ρ (the population correlation)

```
cor(murders$population, murders$total)
```

```
## [1] 0.9635956
```

Assumption:

- Both variables are approximately normally distributed

Pearson correlation coefficient

Null hypothesis: $\rho = 0$

Alternative hypothesis: $\rho \neq 0$

Conclusion: the population correlation is significantly different than 0.

```
cor.test(murders$population, murders$total)
```

```
##
##  Pearson's product-moment correlation
##
## data:  murders$population and murders$total
## t = 25.228, df = 49, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.9367732 0.9791617
## sample estimates:
##       cor
## 0.9635956
```

Spearman rank correlation coefficient

The Spearman correlation coefficient (r_S) is a point estimate of ρ (the population correlation)

- Does not assume both variables are approximately normally distributed
- Nonparametric; based on ranks
- Less sensitive to outliers and skewed distributions

```
cor(murders$population, murders$total, method = "spearman")
```

```
## [1] 0.8906688
```

Spearman rank correlation coefficient

Null hypothesis: $\rho = 0$

Alternative hypothesis: $\rho \neq 0$

Conclusion: the population correlation is significantly different than 0.

```r
cor.test(murders$population, murders$total, method = "spearman")
```

```
##
##  Spearman's rank correlation rho
##
## data:  murders$population and murders$total
## S = 2416.2, p-value < 2.2e-16
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.8906688
```

Example: Prenatal care

In 1989, Seattle and King County in Washington state began a program called First Steps that provides free prenatal care to low-income women.

- Includes both medical attention and advice about lifestyle decisions that may affect birthweight
- The goal of First Steps is to improve birth outcomes, including lower prevalence of low birthweight babies

Example: Prenatal care

`firstep`: indicates participation in First Steps

`gestation`: gestational age at birth (in weeks)

`age`: the mother's age at birth

`wpre`: mother's pre-pregnancy weight in pounds

`wgain`: amount of maternal weight-gain during pregnancy

`education`: years of mother's education

`bwt`: birthweight (grams)

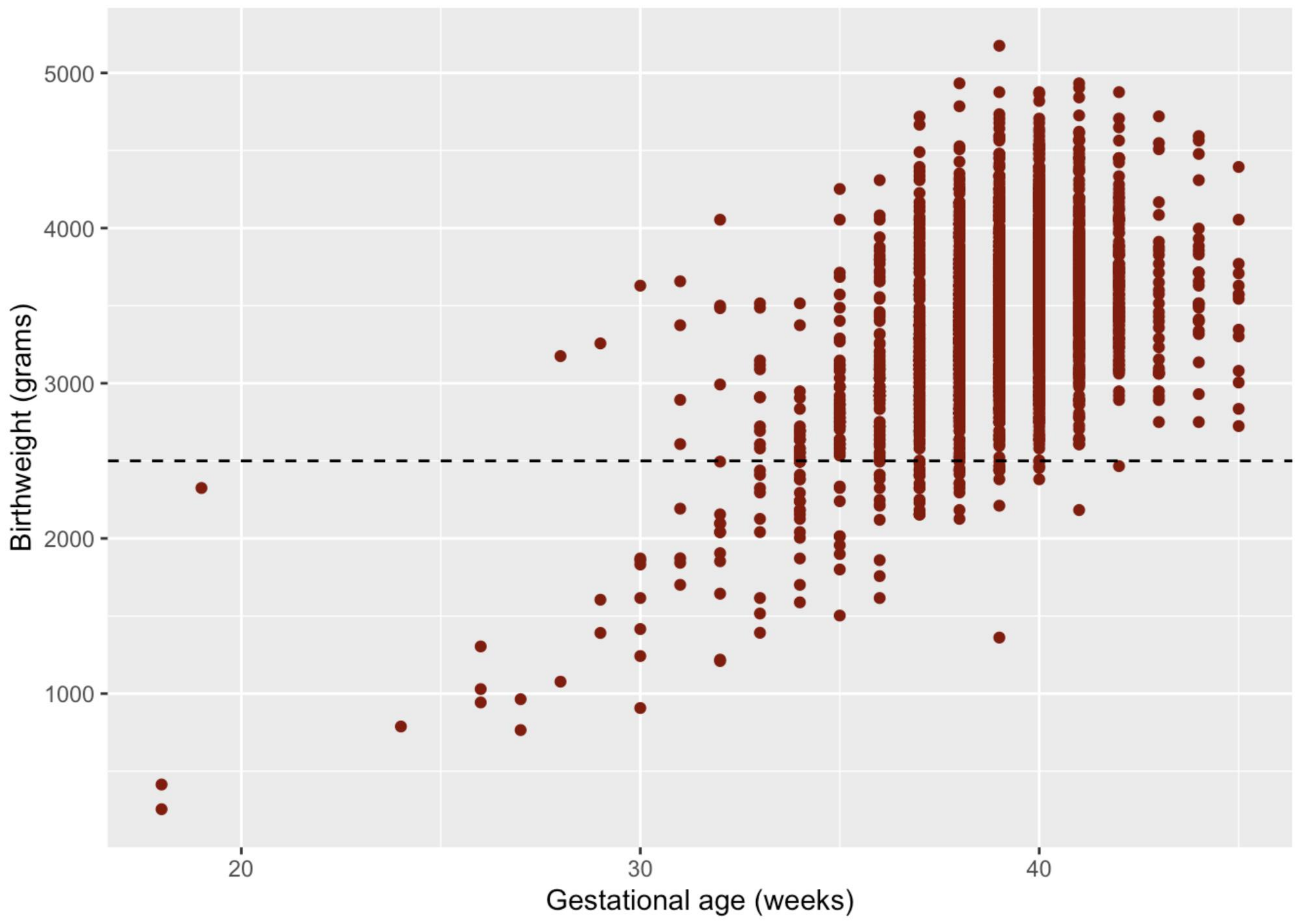
Example: Prenatal care

```r
data <- read.table("KingCounty2001_data.txt", header = TRUE)
head(data)
```

```
##   gender plural age     race parity married  bwt smokeN drinkN firstep welfare
## 1      F      1  31 hispanic      4       1 3118      0      0       1       0
## 2      M      1  23    white      0       1 3466      0      0       1       0
## 3      F      1  24    black      1       0 3147      0      0       1       0
## 4      M      1  26 hispanic      2       1 3969      0      0       1       0
## 5      M      1  34    white      0       0 3005      0      0       1       0
## 6      M      1  15    black      0       0 2920      0      0       1       0
##   smoker drinker wpre wgain education gestation
## 1      N       N  122    22         5        40
## 2      N       N  160    50        12        39
## 3      N       N  150    38        13        40
## 4      N       N  175    15        12        39
## 5      N       N  123    17        17        40
## 6      N       N  180    12         9        36
```

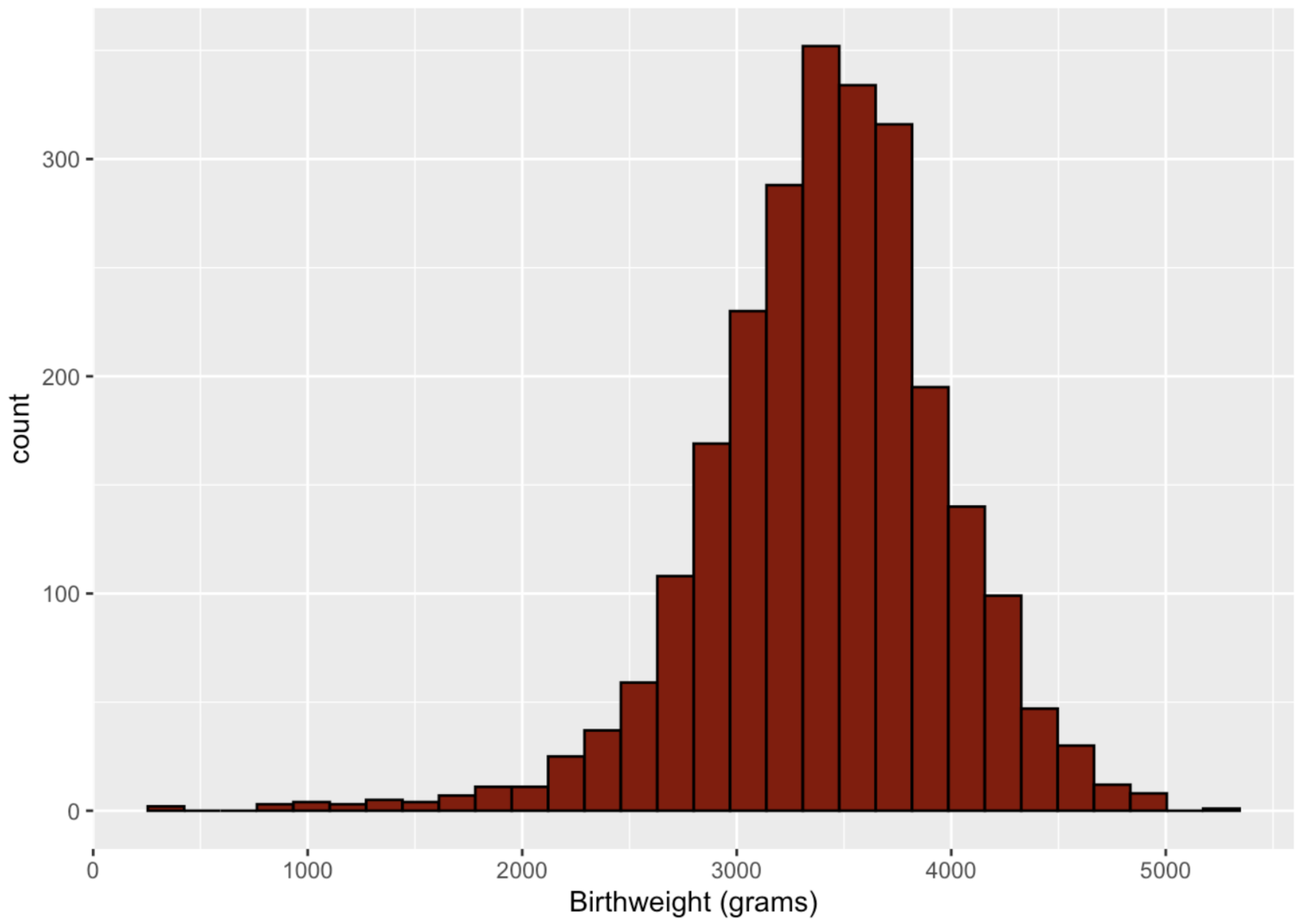
Example: Prenatal care

```r
data %>% ggplot(aes(gestation, bwt)) +
  geom_point(color = "darkred") +
  xlab("Gestational age (weeks)") +
  ylab("Birthweight (grams)") +
  geom_hline(yintercept = 2500, lty = 2)
```

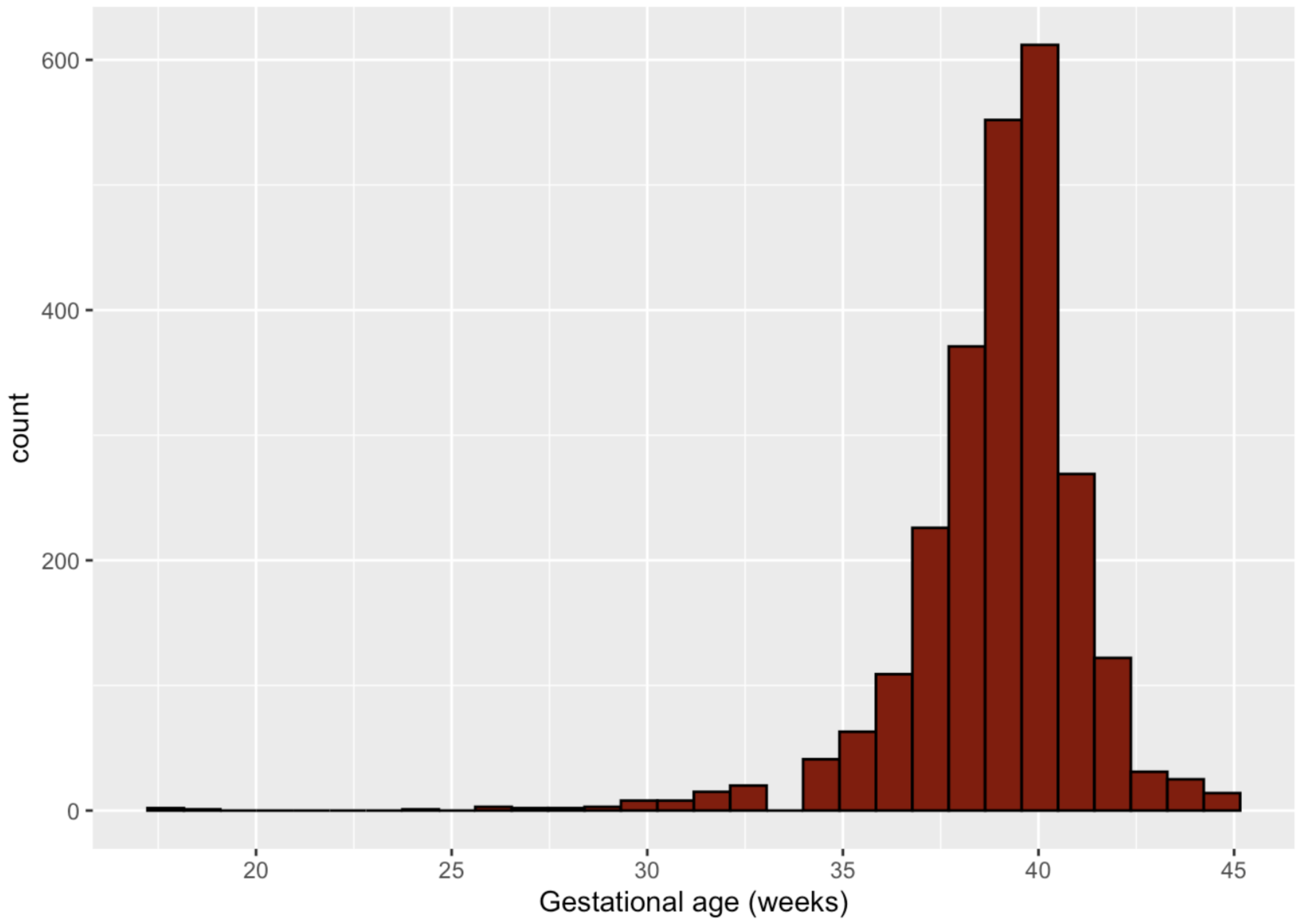


Example: Prenatal care

```
data %>% ggplot(aes(bwt)) +
  geom_histogram(color = "black", fill = "darkred") +
  xlab("Birthweight (grams)")
```




```
data %>% ggplot(aes(gestation)) +
  geom_histogram(color = "black", fill = "darkred") +
  xlab("Gestational age (weeks)")
```



Example: Prenatal care

```r
cor(data$gestation, data$bwt)
```

```
## [1] 0.527407
```

```r
cor.test(data$gestation, data$bwt)
```

```
##
##  Pearson's product-moment correlation
##
## data:  data$gestation and data$bwt
## t = 31.026, df = 2498, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.4985114 0.5551319
## sample estimates:
##      cor
## 0.527407
```

```r
cor(data$gestation, data$bwt, method = "spearman")
```

```
## [1] 0.4093581
```

```r
cor.test(data$gestation, data$bwt, method = "spearman")
```

```
##
##  Spearman's rank correlation rho
##
## data:  data$gestation and data$bwt
## S = 1538129750, p-value < 2.2e-16
## alternative hypothesis: true rho is not equal to 0
## sample estimates:
##       rho
## 0.4093581
```

Summary

- Correlation quantifies the linear relationship between two continuous variables
- The Pearson correlation coefficient *r* is sensitive to outliers and requires the two continuous variables of interest to be approximately normally distributed
- If one or both variables are not approximately normally distributed the Spearman correlation r_s should be used instead

Linear Regression I

Simple linear regression

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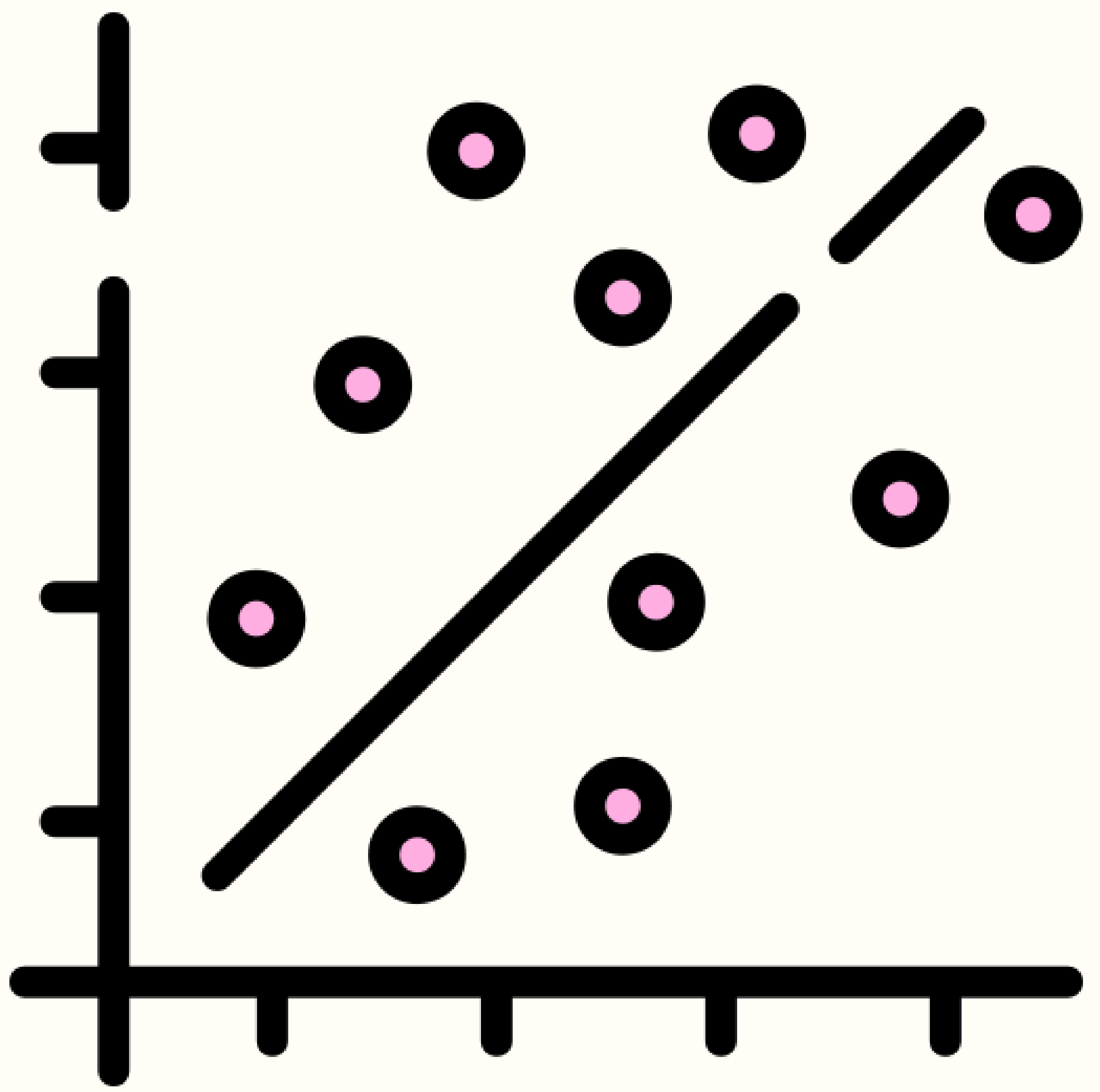
Linear regression

Like correlation analysis, linear regression can be used to investigate the linear relationship between two continuous random variables.

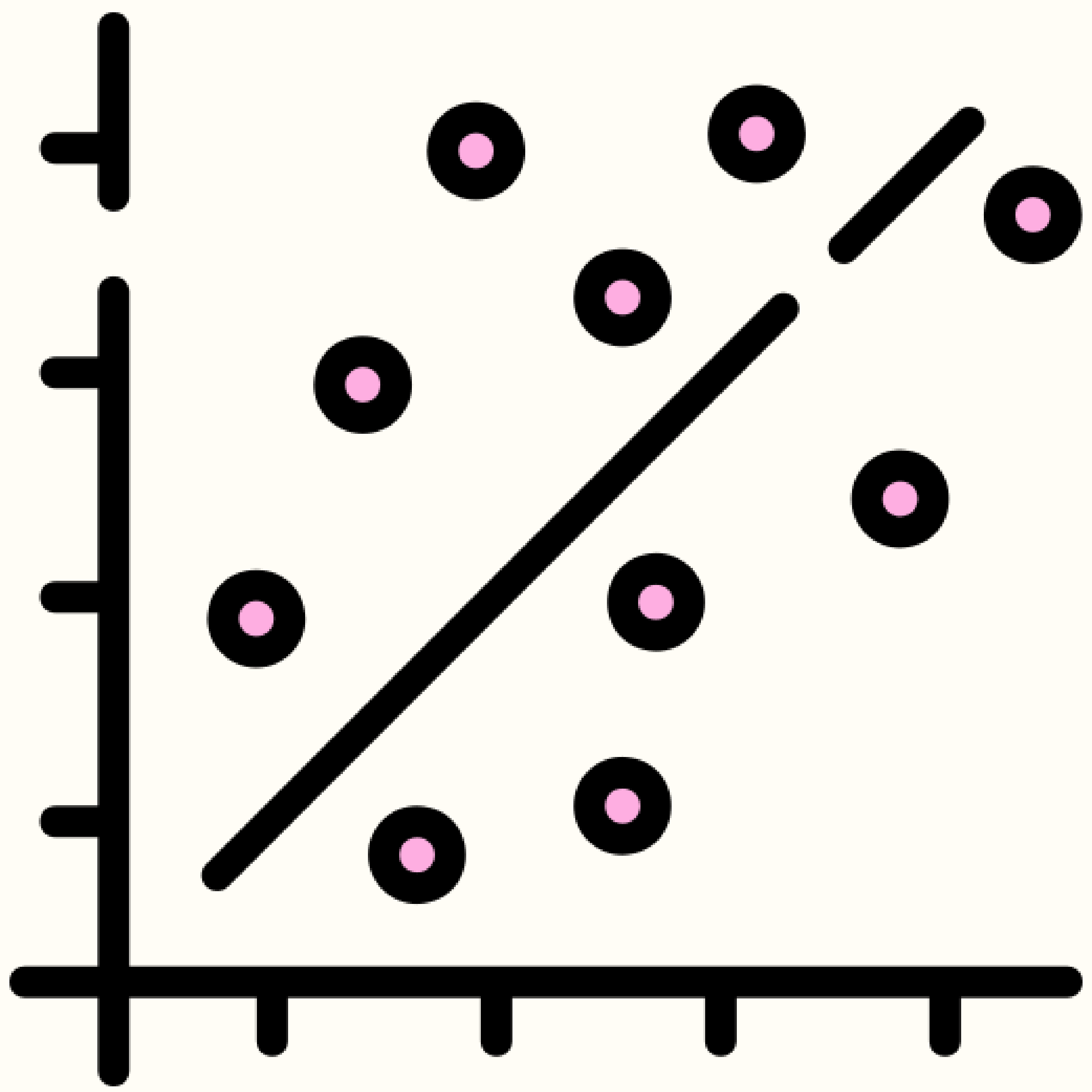
- Relationships between a continuous outcome variable and one or more risk factors that can be continuous, discrete, or categorical (nominal or ordinal), or some combination of these.



Linear regression

Linear regression can be used to

- Estimate or predict the value of a continuous outcome measurement given a set of risk factors
- Adjust for confounders when examining the relationship between an exposure and a continuous outcome
- Test for effect measure modification



Simple linear regression

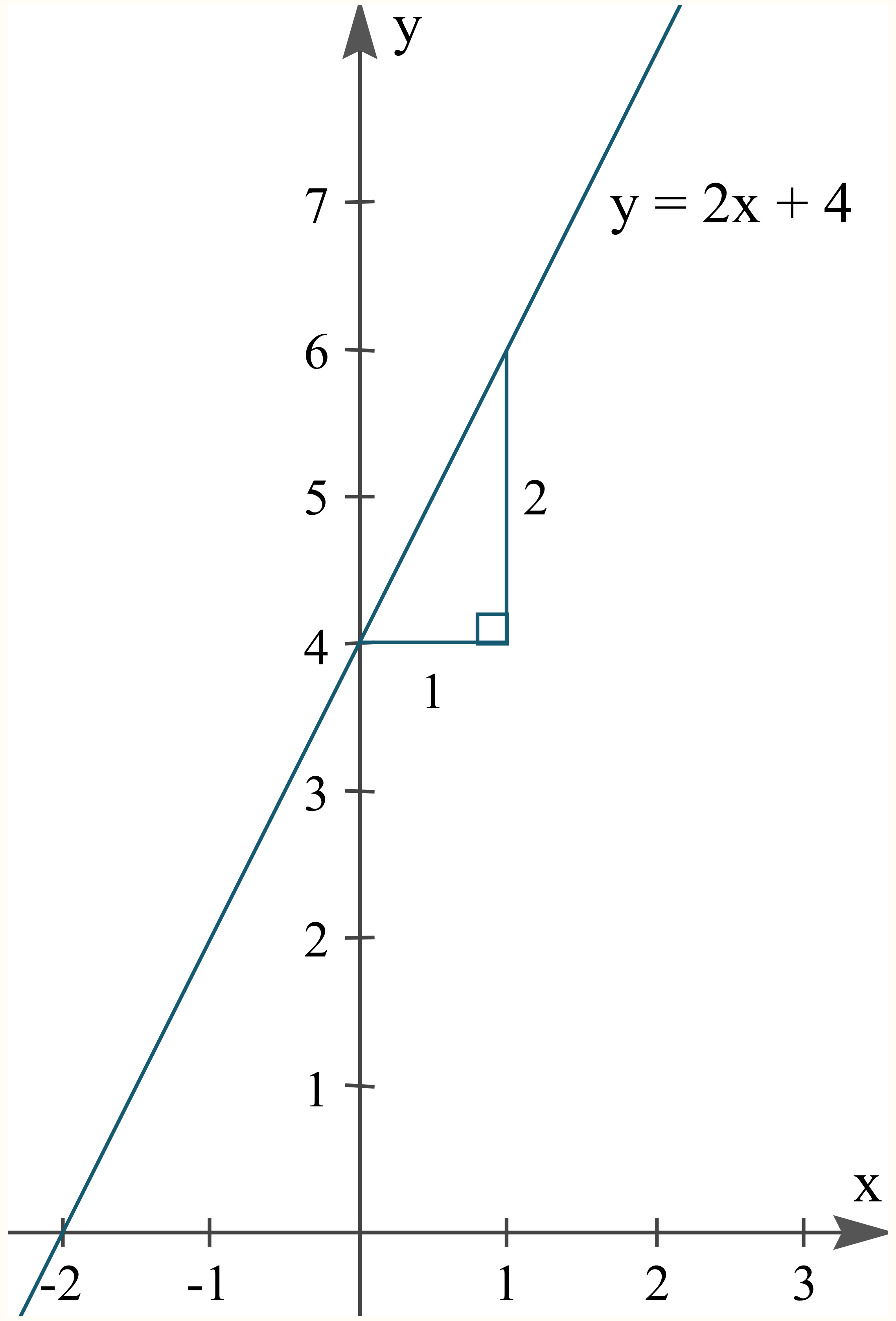
Simple linear regression can be used to explore the nature of the relationship between two continuous variables

- Regression looks at the change in one variable (outcome or response or dependent variable) that corresponds to a given change in the other variable (explanatory variable or exposure or predictor or covariate or independent variable)

$y = mx + b$

m: slope

b: y intercept



Assumptions

1. Linearity:
   - The relationship between $\mu_{y|x}$ (mean of *y* given *x*) and *x* is linear
2. Normality:
   - For a given value of *x*, *y* is normally distributed
3. Homoscedasticity:
   - The standard deviation $\sigma_{y|x}$ is constant for all *x*
4. Independence:
   - We have *n* independent pairs of observations

Example: Prenatal care

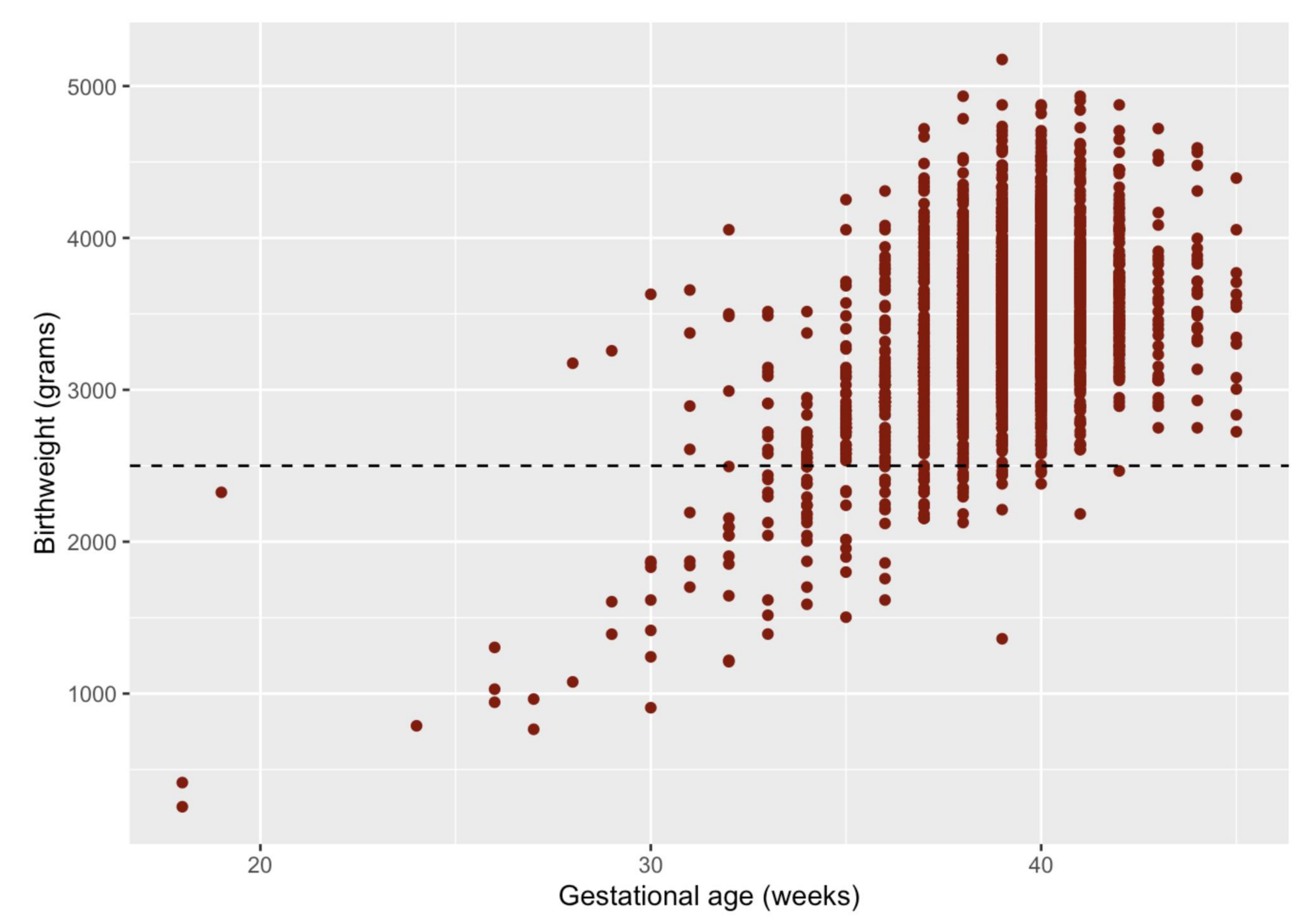
```r
data <- read.table("KingCounty2001_data.txt", header = TRUE)
head(data)
```

```
##   gender plural age     race parity married  bwt smokeN drinkN firstep welfare
## 1      F      1  31 hispanic      4       1 3118      0      0       1       0
## 2      M      1  23    white      0       1 3466      0      0       1       0
## 3      F      1  24    black      1       0 3147      0      0       1       0
## 4      M      1  26 hispanic      2       1 3969      0      0       1       0
## 5      M      1  34    white      0       0 3005      0      0       1       0
## 6      M      1  15    black      0       0 2920      0      0       1       0
##   smoker drinker wpre wgain education gestation
## 1      N       N  122    22         5        40
## 2      N       N  160    50        12        39
## 3      N       N  150    38        13        40
## 4      N       N  175    15        12        39
## 5      N       N  123    17        17        40
## 6      N       N  180    12         9        36
```

Example: Prenatal care

Suppose the outcome is birthweight in grams and the exposure is gestational age in weeks.

- A linear regression model will quantify how birthweight and gestational age are associated



The `lm` function

The **`lm`** function stands for linear model and is used to fit a linear regression model

- The primary inputs are formula and data
- Simple linear regression formula: **`outcome ~ predictor`**

```
m1 <- lm(bwt ~ gestation, data = data)
```

Regression line equation:

$$\mu_{y|x} = \beta_1 x + \beta_0$$

$$\mu_{bwt|gestation} = \beta_1(\text{gestation}) + \beta_0$$

The `lm` function

The `lm` function stands for linear model and is used to fit a linear regression model

- The summary function outputs details about the fit model

Fitted regression line equation:

Mean birthweight = 124.1(gestation) – 1410.7

```
summary(m1)
```

```
##
## Call:
## lm(formula = bwt ~ gestation, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2068.46  -301.93    -9.56   273.44  1745.54
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -1410.7      155.8  -9.055   <2e-16 ***
## gestation      124.1        4.0  31.026   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 475.3 on 2498 degrees of freedom
## Multiple R-squared:  0.2782, Adjusted R-squared:  0.2779
## F-statistic: 962.6 on 1 and 2498 DF,  p-value: < 2.2e-16
```

Coefficient interpretation

Fitted regression line equation:

Mean birthweight = 124.1(gestation) – 1410.7

Slope interpretation: for every 1 week increase in gestational age, birthweight increases by 124.1 grams on average.

Intercept interpretation: the estimated mean birthweight when gestational age is 0 weeks is -1410.7 grams.

```
summary(m1)
```

```
##
## Call:
## lm(formula = bwt ~ gestation, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2068.46  -301.93    -9.56   273.44  1745.54
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -1410.7      155.8  -9.055   <2e-16 ***
## gestation      124.1        4.0  31.026   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 475.3 on 2498 degrees of freedom
## Multiple R-squared:  0.2782, Adjusted R-squared:  0.2779
## F-statistic: 962.6 on 1 and 2498 DF,  p-value: < 2.2e-16
```

Coefficient of determination

R^2 is the coefficient of determination, which quantifies the proportion of the variance in the dependent variable that is explained by the independent variable.

- Ranges from 0 to 1
- A higher R^2 value indicates a better fit model
- In simple linear regression, R^2 is the Pearson correlation coefficient squared

```r
summary(m1)
```

```
##
## Call:
## lm(formula = bwt ~ gestation, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2068.46  -301.93    -9.56   273.44  1745.54
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -1410.7      155.8  -9.055   <2e-16 ***
## gestation      124.1        4.0  31.026   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 475.3 on 2498 degrees of freedom
## Multiple R-squared:  0.2782, Adjusted R-squared:  0.2779
## F-statistic: 962.6 on 1 and 2498 DF,  p-value: < 2.2e-16
```

Confidence interval for slope

The confidence interval for all of the coefficients of the model, including the intercept, can be calculated using the `confint` function

```
confint(m1)
```

```
##                   2.5 %     97.5 %
## (Intercept) -1716.1938 -1105.1835
## gestation     116.2624   131.9502
```

Hypothesis tests

The p-value of the slope coefficient is used to test if the relationship between the outcome and exposure is significant

- If the p-value is < 0.05, the slope is not equal to 0 and there is a statistically significant relationship between the outcome and exposure

```
summary(m1)
```

```
##
## Call:
## lm(formula = bwt ~ gestation, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2068.46  -301.93    -9.56   273.44  1745.54
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -1410.7      155.8  -9.055   <2e-16 ***
## gestation      124.1        4.0  31.026   <2e-16 ***
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## F-statistic: 962.6 on 1 and 2498 DF,  p-value: < 2.2e-16
```

Prediction

One important use of a linear regression model is prediction of a continuous outcome

- Example: the predicted birthweight of a baby with gestational age equal to 40 weeks is 124.1(40) – 1410.7 = 3,553.3 grams.

```
summary(m1)
```

```
##
## Call:
## lm(formula = bwt ~ gestation, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2068.46  -301.93    -9.56   273.44  1745.54
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -1410.7      155.8  -9.055   <2e-16 ***
## gestation      124.1        4.0  31.026   <2e-16 ***
## ---
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##
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## Multiple R-squared:  0.2782, Adjusted R-squared:  0.2779
## F-statistic: 962.6 on 1 and 2498 DF,  p-value: < 2.2e-16
```

Prediction

One important use of a linear regression model is prediction of a continuous outcome

- Example: the predicted birthweight of a baby with gestational age equal to 28 weeks is 124.1(28) – 1410.7 = 2,064.1 grams.

```
summary(m1)
```

```
##
## Call:
## lm(formula = bwt ~ gestation, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2068.46  -301.93    -9.56   273.44  1745.54
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -1410.7      155.8  -9.055   <2e-16 ***
## gestation      124.1        4.0  31.026   <2e-16 ***
## ---
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##
## Residual standard error: 475.3 on 2498 degrees of freedom
## Multiple R-squared:  0.2782, Adjusted R-squared:  0.2779
## F-statistic: 962.6 on 1 and 2498 DF,  p-value: < 2.2e-16
```
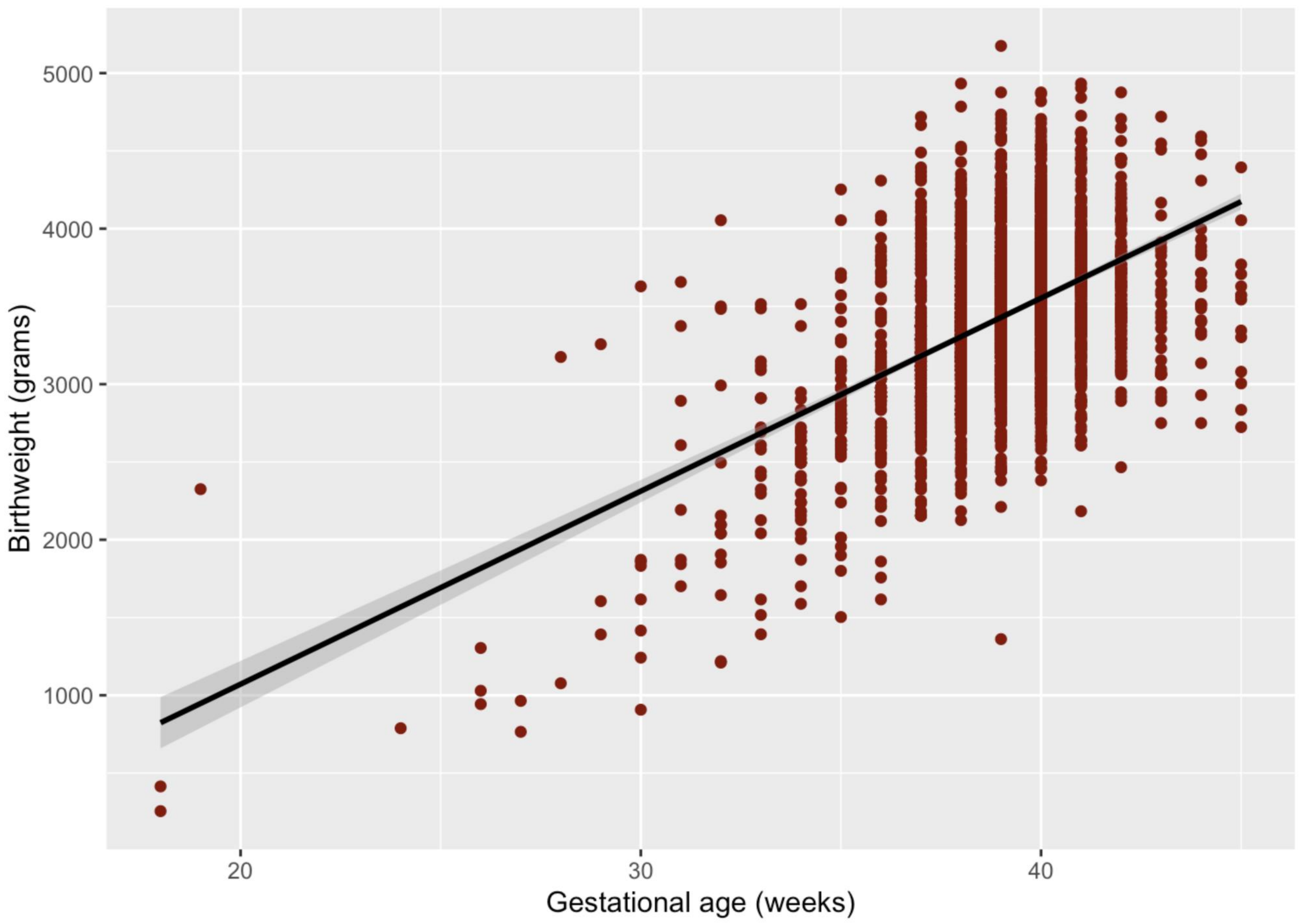

Regression line plot

```
data %>%
  ggplot(aes(gestation, bwt)) +
  geom_point(color = "darkred") +
  geom_smooth(method = "lm", se = TRUE, color = "black") +
  xlab("Gestational age (weeks)") +
  ylab("Birthweight (grams)")
```



Summary

- Simple linear regression quantifies the relationship between two continuous variables
- The `lm` function can be used to fit a linear regression model
- The `summary` function outputs details about the model, including coefficient values and R^2
- A linear regression model can be used for prediction
- A plot of the fitted regression line can be created using the `geom_smooth` geometry

Linear Regression I

Evaluation of model assumptions

Heather Mattie, PhD

Linear regression assumptions

1. Linearity:
   - The relationship between $\mu_{y|x}$ (mean of *y* given *x*) and *x* is a straight line
2. Normality:
   - For a given value of *x, y* is normally distributed
3. Homoscedasticity:
   - The standard deviation $\sigma_{y|x}$ is constant for all *x*
4. Independence:
   - We have *n* independent pairs of observations

Example: Prenatal care

```
data <- read.table("KingCounty2001_data.txt", header = TRUE)
head(data)
```

```
##   gender plural age     race parity married  bwt smokeN drinkN firstep welfare
## 1      F      1  31 hispanic      4       1 3118      0      0       1       0
## 2      M      1  23    white      0       1 3466      0      0       1       0
## 3      F      1  24    black      1       0 3147      0      0       1       0
## 4      M      1  26 hispanic      2       1 3969      0      0       1       0
## 5      M      1  34    white      0       0 3005      0      0       1       0
## 6      M      1  15    black      0       0 2920      0      0       1       0
##   smoker drinker wpre wgain education gestation
## 1      N       N  122    22         5        40
## 2      N       N  160    50        12        39
## 3      N       N  150    38        13        40
## 4      N       N  175    15        12        39
## 5      N       N  123    17        17        40
## 6      N       N  180    12         9        36
```

ggforitfy and autoplot

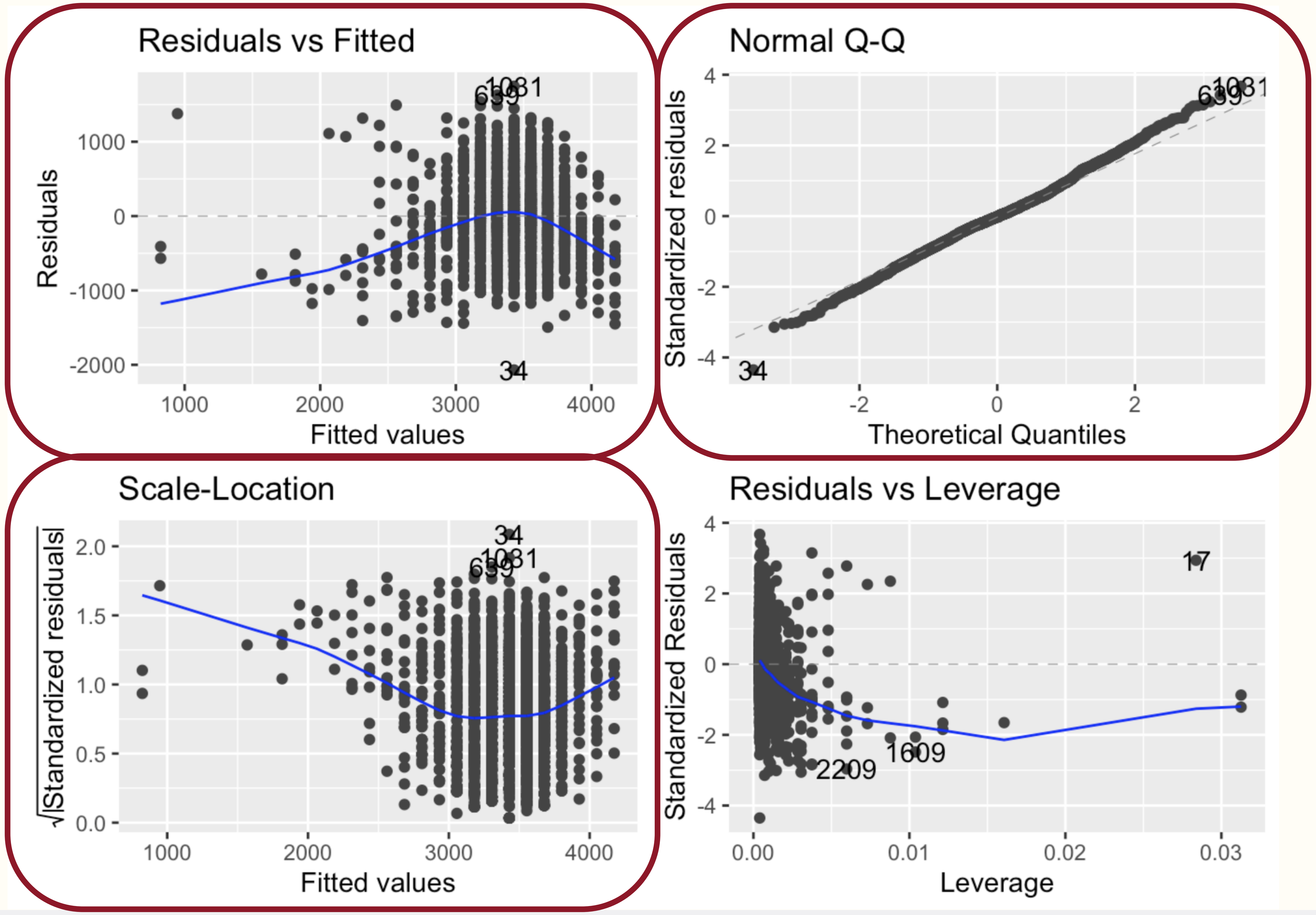
The function **autoplot** from the **ggfortify** package can be used to create plots to check if the assumptions of linear regression are met

```
library(ggfortify)
autoplot(m1)
```

ggforitfy and autoplot

The function `autoplot` from the `ggfortify` package can be used to create plots to check if the assumptions of linear regression are met

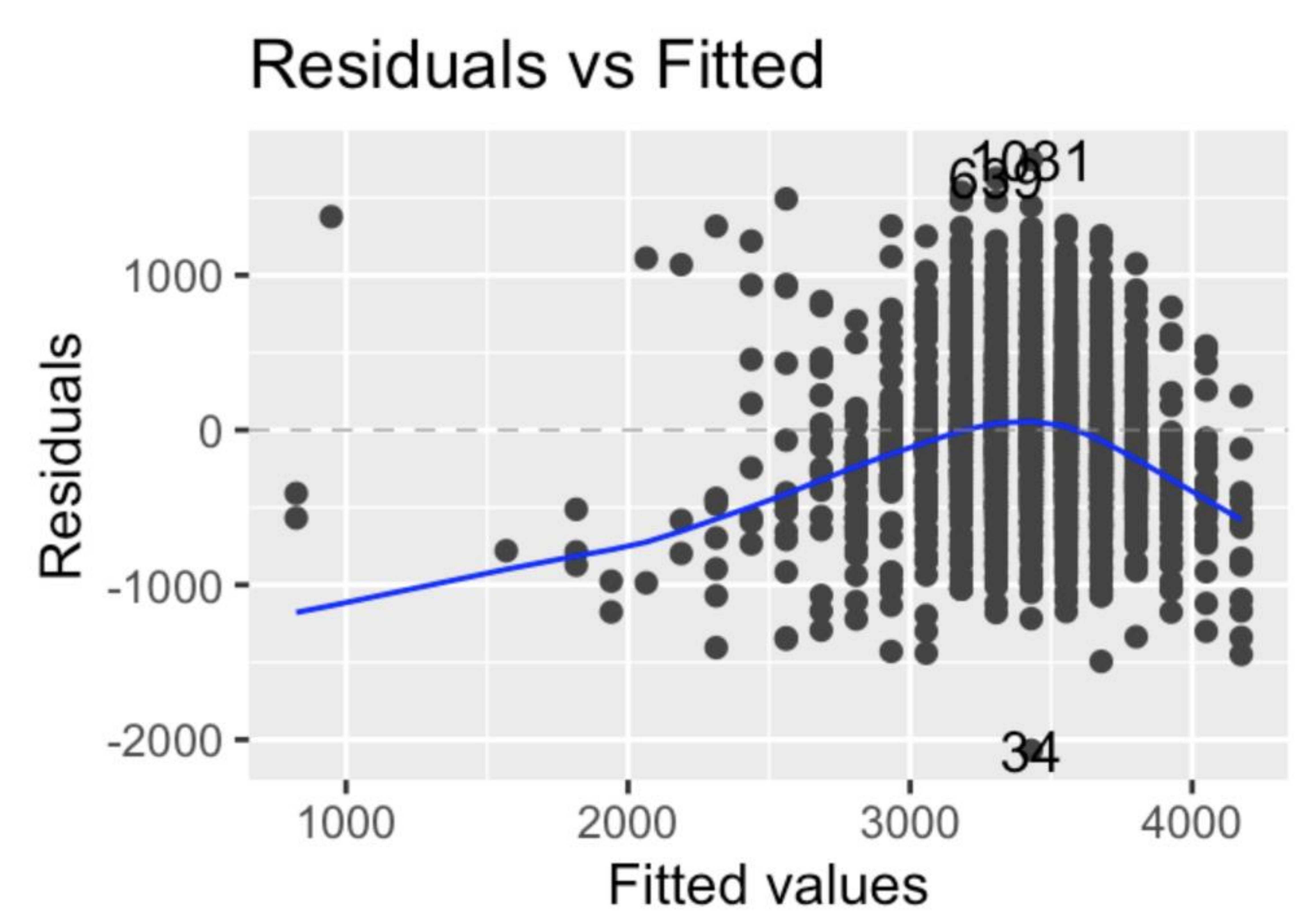
- Four plots are created, we will use 3 of those plots



Linearity

A plot of the residuals (difference between the actual values and the values predicted with the regression line) versus the fitted values is used to check linearity.

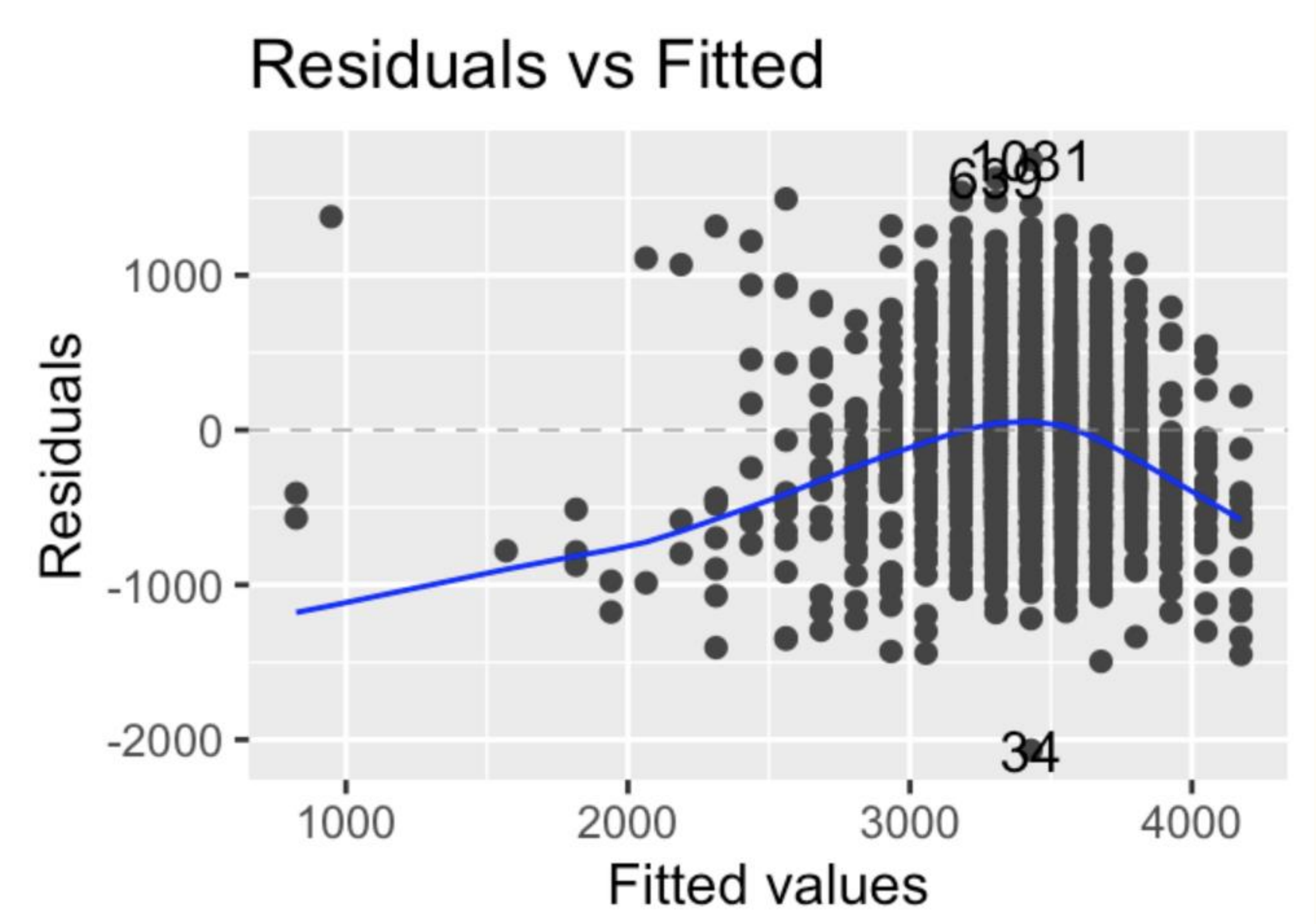
- A horizontal line, without distinct patterns is an indication for a linear relationship



Linearity

The linearity assumption is violated in our dataset

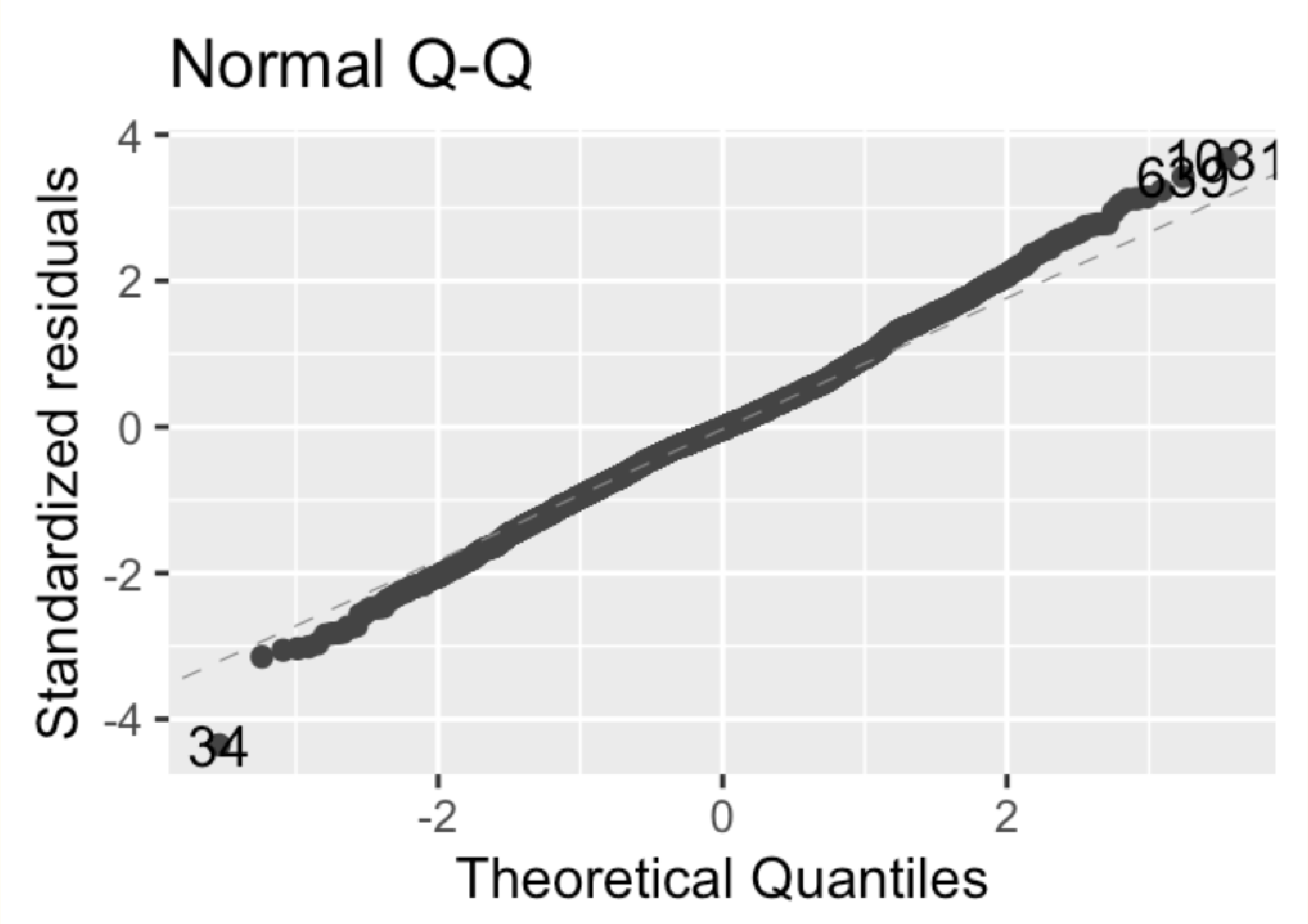
- Due to the relatively small amount of data with lower values of gestational age
- A potential solution would be to transform the exposure variable (log, square root, etc.)



Normality

A quantile-quantile plot (Normal Q-Q plot) is used to check the normality assumption.

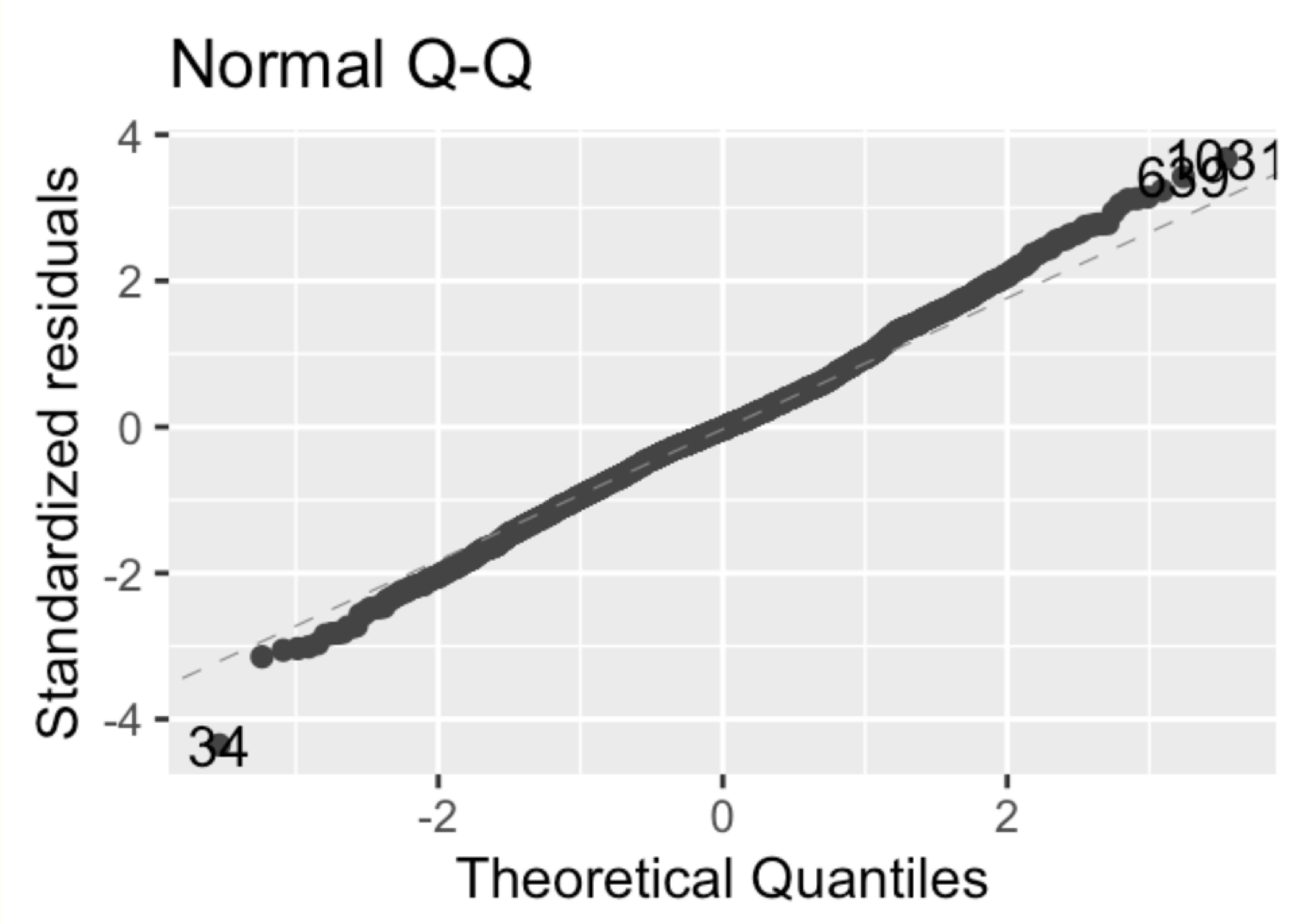
- The normality assumption is met if the standardized residuals follow the straight dashed line



Normality

Looking at the plot, it's difficult to be sure the normality assumption is met.

- Can use the Shapiro-Wilk test to test the normality assumption



Normality

The normality assumption is not met in the dataset since the p-value is < 0.05.

A few potential solutions include

- Applying data transformations to the exposure variable
- Using robust regression methods that are less sensitive to non-normal data or outliers
- Remove outliers

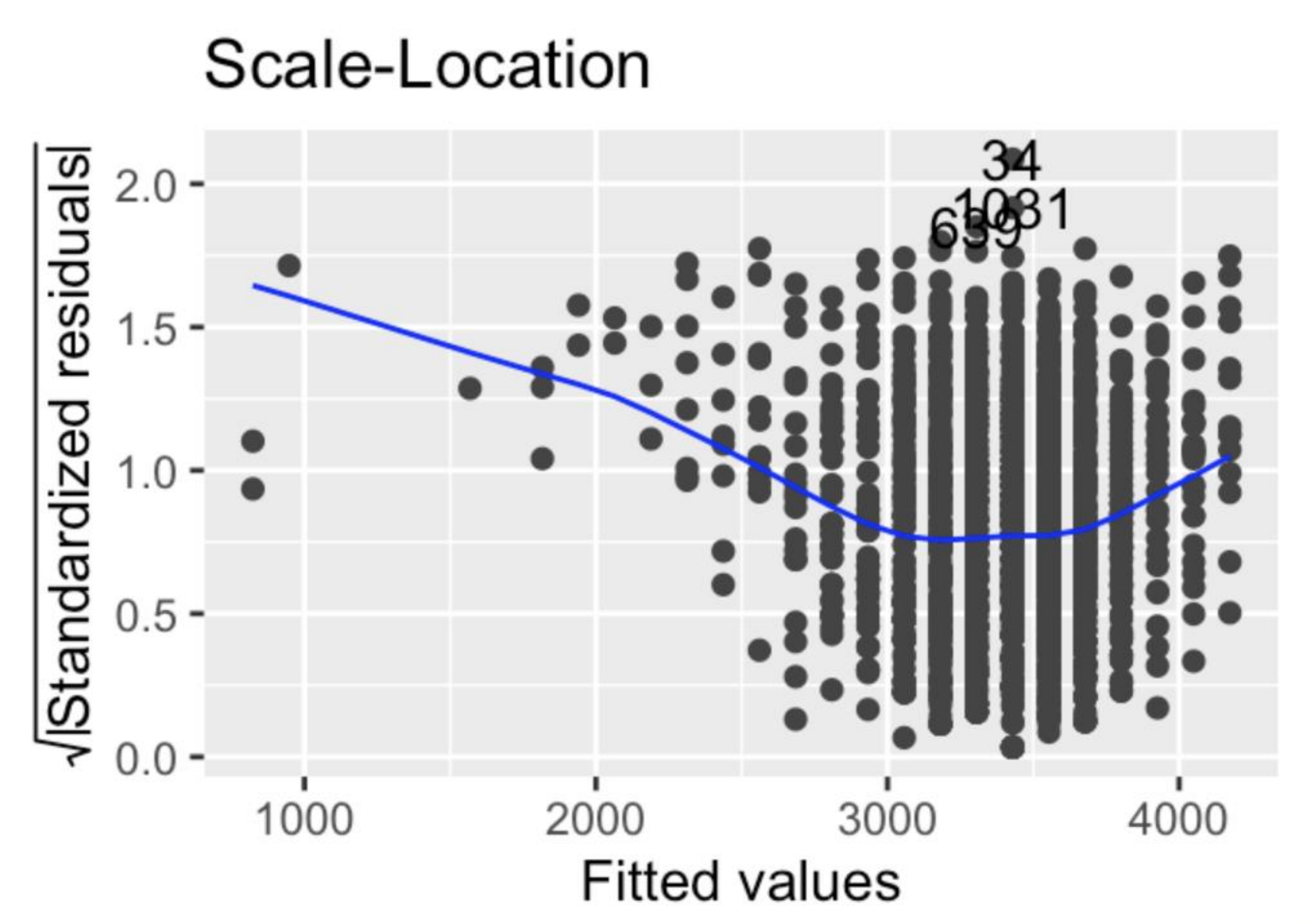
```
shapiro.test(residuals(m1))
```

```
##
##  Shapiro-Wilk normality test
##
## data:  residuals(m1)
## W = 0.99666, p-value = 2.551e-05
```

Homoscedasticity

A plot of the square root of the standardized residuals versus the fitted values is used to check the homoscedasticity assumption.

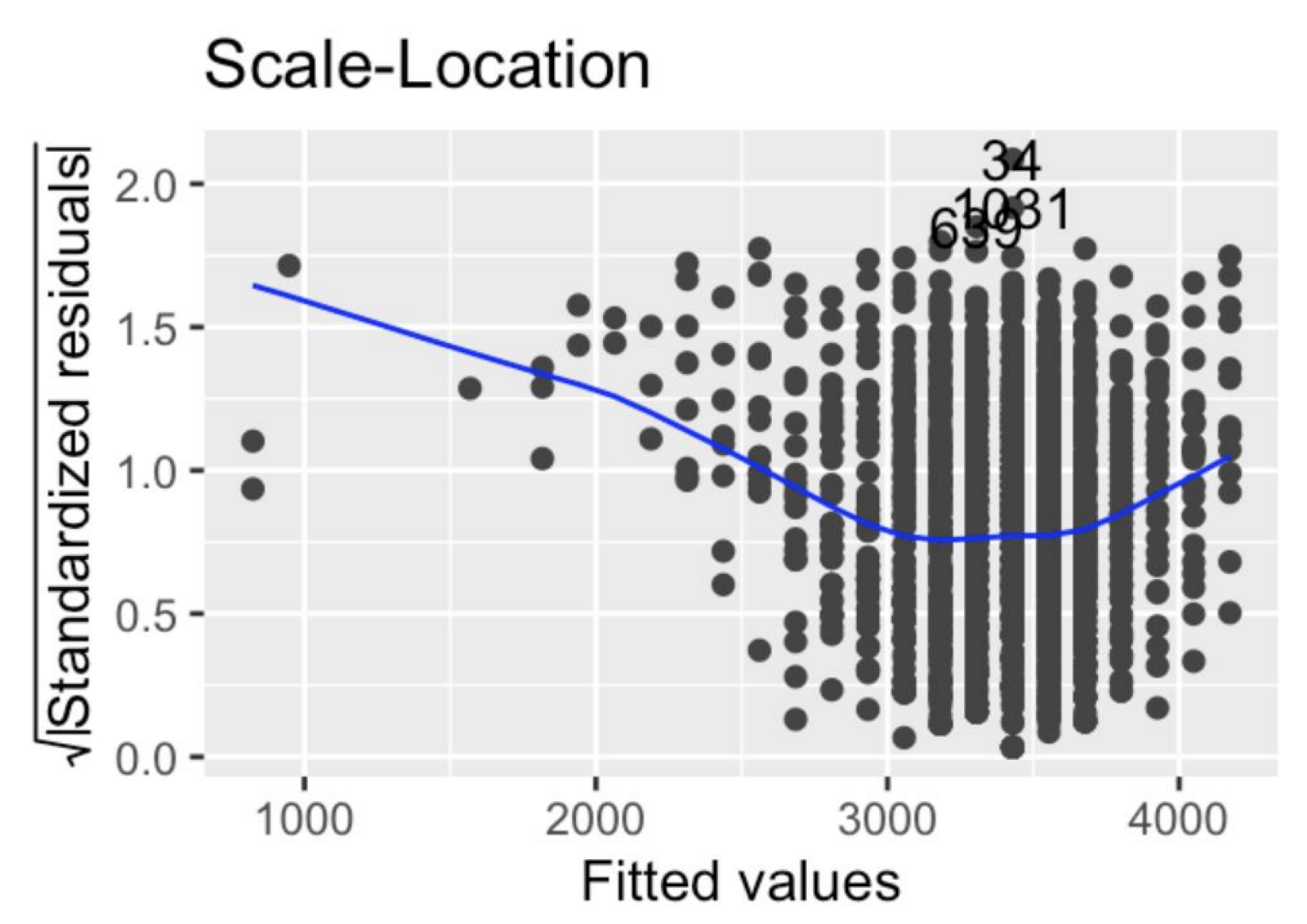
- Called a scale-location or spread-location plot
- A horizontal line with equally spread points is a good indication of homoscedasticity.



Homoscedasticity

The homoscedasticity assumption is violated in our dataset.

- Due to the relatively small amount of data with lower values of gestational age
- A potential solution would be to transform the outcome variable (log, square root, etc.)



Independence

The only way to check the independence assumption is having knowledge of the study / sampling design.

- Contact those responsible for data collection
- Read study documentation
- Create a subset of the original dataset that only includes independent pairs of observations

The independence assumption holds for this dataset.

Summary

- The assumptions of linear regression should be checked before choosing and interpreting a final model
- The linearity assumption can be checked with a plot of the residuals versus the fitted values
- The normality assumption can be checked with a Normal Q-Q plot or Shapiro-Wilk test
- The homoscedasticity assumption can be checked with a plot of the square root of the standardized residuals versus the fitted values
- The independence assumption can only be checked by verifying the study / sampling design for data collection